HYPOTHESES OF THE SPLIT-EXACTNESS LEMMA HOLD FOR POINTED CURVES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1146-1152 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1154-1167, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_pic.json (pb=710, stmt 1146-1152, proof 1154-1167).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 02KH: sources/stacks-pic/pinned-original-source.tex, source lines 1167–1167. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9N: sources/stacks-pic/pinned-original-source.tex, source lines 1028–1038. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04L2: sources/stacks-pic/pinned-original-source.tex, source lines 1167–1167. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let X be a smooth projective curve over k which has a k-rational point. Then the hypotheses of Lemma 0B9N are satisfied.*

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The meaning of the phrase "has a k-rational point" is exactly that the structure morphism $f : X \to \mathrm{Spec}(k)$ has a section, which verifies the first condition. By Varieties, Lemma 04L2 we see that $k' = H^0(X, \mathcal{O}_X)$ is a field extension of k. Since X has a k-rational point there is a k-algebra homomorphism $k' \to k$ and we conclude $k' = k$. Since k is a field, any morphism $T \to \mathrm{Spec}(k)$ is flat. Hence we see by cohomology and base change (Cohomology of Schemes, Lemma 02KH) that $\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism. This finishes the proof. □